

2.2 Cell Metabolism II

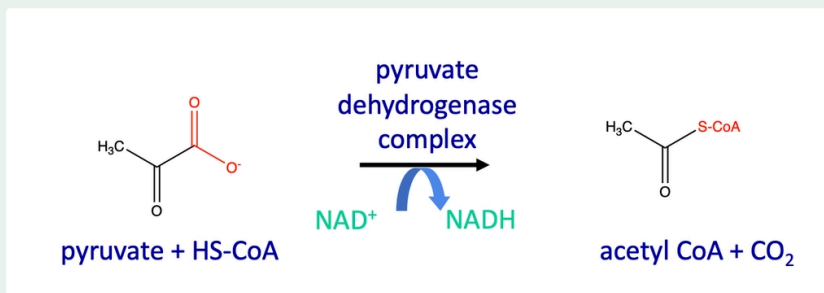
▼ Learning Objective

A session that will explore the entry of acetyl CoA into the TCA Cycle after its production from pyruvate in glycolysis

To understand what happens in the TCA Cycle

1.13-50

▼ Describe how pyruvate is converted to acetyl CoA.



- $\text{NAD}^+ \rightarrow \text{NADH}$
- HS-CoA is just **coenzyme A** with Sulfur and Hydrogen
- The **pyruvate dehydrogenase complex** catalyses this reaction

▼ What happens when acetyl CoA (2C) enters the mitochondrial matrix for the TCA Cycle?

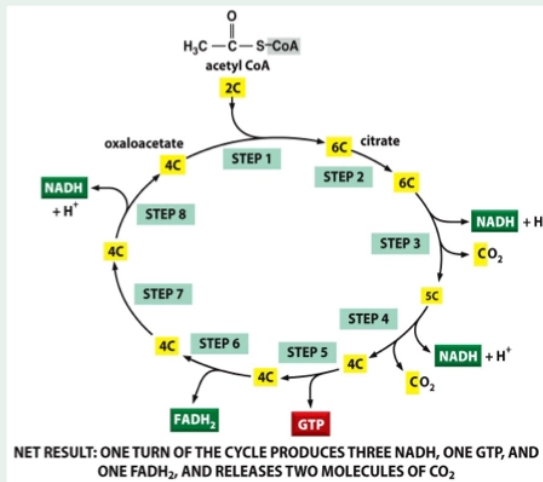
- It reacts with **oxaloacetate (4C)** to produce **citrate (6C)**

▼ Describe briefly what happens in the TCA Cycle.

- **Acetyl CoA (2C)** enters from the Link Reaction and reacts with **oxaloacetate (4C)** to form **citrate (6C)**
- **Redox and decarboxylation reactions** occur which produce **NADH , CO_2 , GTP and FADH_2** molecules using the citrate and convert it **back to oxaloacetate (4C)**
- This **oxaloacetate (4C)** has now been **reproduced** and can be used in another round of the TCA Cycle.

▼ Describe the molecules that are produced via one TCA Cycle.

- 2x CO₂ produced from **citrate (6C) → oxaloacetate (4C)**
- 3x NADH from different reactions (2 arise alongside CO₂ removals)
- 1x FADH₂ from 4C to 4C reaction
- 1x GTP from 4C to 4C reaction



▼ How much ATP (equivalents) can hence be produced via one TCA Cycle?

- 12x ATP.

▼ How does the TCA Cycle help to produce energy, if no ATP is even produced?

- The **reduced coenzymes (3x NADH, 1x FADH₂)** that are produced in the TCA Cycle help to produce the ATP in oxidative phosphorylation
 - They **supply the H⁺ and e⁻ used in the electron transport chain**

▼ What stops the TCA Cycle from occurring under anaerobic conditions?

- The TCA Cycle requires oxidised coenzymes (NAD⁺/FAD) for it to proceed, so it can reduce them to NADH/FADH₂ during the cycle
- Since these oxidised coenzymes can only be produced via their reoxidation during oxidative phosphorylation (where oxygen accepts their electrons), a lack of oxygen would mean the coenzymes continue to exist in their reduced state
- Hence the TCA Cycle requires oxygen to go ahead.

▼ Where are the enzymes used in the TCA Cycle stored?

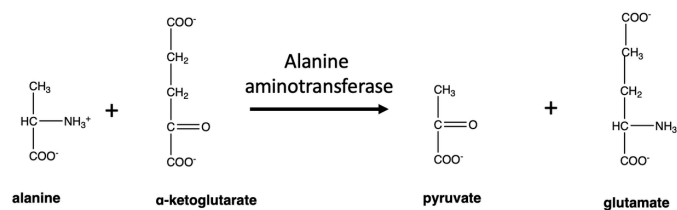
- In the mitochondrial matrix as they are **soluble proteins**.

- ▼ What enzyme is the exception to this rule and where is it found?
 - **Succinate dehydrogenase.**
 - This enzyme is used to form complex II of the electron transport chain and so is found on the **inner mitochondrial membrane**
- ▼ How do amino acids enter the TCA Cycle?
 - **Proteins** are down into **amino acids**
 - The **carbon skeletons** of these amino acids can form **acetyl CoA** or **other intermediates present in the TCA Cycle**
 - The intermediates formed can then either be used to make the same products from the **TCA Cycle** or in **gluconeogenesis (reverse of TCA Cycle)**
- ▼ What does the term glucogenic imply?
 - The amino acids that form TCA Cycle intermediates can produce either glucose or ketone bodies
 - **Those that produce glucose** are described as **glucogenic**
- ▼ What does the term ketogenic imply?
 - The amino acids that form TCA Cycle intermediates can produce either glucose or ketone bodies (i.e acetyl CoA)
 - **Those that produce ketone bodies** are described as **ketogenic**
- ▼ What is a transamination reaction?
 - A reaction that involves the **transfer of an amine group** from **one molecule to another.**
- ▼ How is a transamination reaction useful for amino acid metabolism?
 - Transferring amine groups from amino acids to keto acids **allows amino acids to be metabolised without harm to the cell**
 - This is because **metabolising amine groups can be toxic to humans**
 - Overall, it **helps us to metabolise amino acids efficiently.**
- ▼ What are aminotransferases?
 - Enzymes that are involved in the **catalysis of transamination reactions.**
- ▼ Which two molecules are involved in transamination reactions for amino acids?
 - **Amino acids and ketoacids**

▼ How is urea produced in protein metabolism?

- When amino acids are metabolised, only the **carbon skeleton** is productive (gives us energy)
- The **amine group is hence not metabolised** into the TCA Cycle, but is instead **transferred to the ketoacid** via transamination, while the **carbon skeleton of the amino acid produces an intermediate**
- The "other molecule" hence has an amine group on it but gets rid of it and as a result, urea is produced via the **urea cycle**
- The diagram below explains this well, using alanine as an example.

Alanine metabolism as an exemplar



Alanine (C3) undergoes transamination by the action of the enzyme alanine aminotransferase.

- Pyruvate can undergo decarboxylation and the acetyl CoA generated can enter the TCA cycle.
- Glutamate is re-converted to α -ketoglutarate by glutamate dehydrogenase.
- This generates NH_4^+ which is ultimately converted to urea.

▼ What is the difference between a ketoacid and an amino acid?

- A ketoacid has a ketone functional group
- An amino acid has an amine functional group

▼ What is the role of the glycerol-phosphate shuttle?

- Its role is to **transfer NADH (electrons)** from the **cytosol into the mitochondrial matrix** so that it can be **reoxidised in oxidative phosphorylation**
- It does this for **skeletal muscle and the brain**
- **NOTE:** this is done for NADH only and not FADH₂ because only glycolysis products have this issue (of going from cytoplasm to mitochondrial matrix)

▼ Describe the function of the glycerol-phosphate shuttle.

- **NADH** adds its electrons to **Dihydroxyacetone phosphate (DHAP)**, converting it to **glycerol-3-phosphate** and **NADH** becomes oxidised to **NAD⁺**
 - This reaction is catalysed by **cytoplasmic glycerol-3-phosphate dehydrogenase**
- **Glycerol-3-phosphate** adds these electrons back to **FAD**, reducing it to **FADH₂** and converting itself back to **Dihydroxyacetone phosphate (DHAP)**
 - This reaction is catalysed by **mitochondrial glycerol-3-phosphate dehydrogenase**
- **FADH₂** then passes its electrons onto **coenzyme Q** (ubiquinone - via complex II in the electron transport chain) where they can be used in oxidative phosphorylation

The Glycerol phosphate shuttle

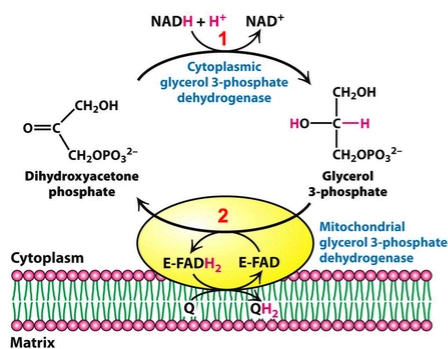


Figure 18-34
Biochemistry, Seventh Edition
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Electrons from **NADH**, rather than **NADH** itself are carried across the mitochondrial membrane via a **shuttle**.

1. Cytosolic glycerol 3-phosphate dehydrogenase transfers electrons from **NADH** to DHAP to generate glycerol 3-phosphate.
2. A membrane bound form of the same enzyme transfers the electrons to **FAD**. These then get passed to co-enzyme Q, part of the electron transport chain.
MBC - Cell integrity

▼ Describe the path that the electrons from NADH take through the glycerol-phosphate shuttle.

- Electrons are transferred from **NADH** to **DHAP** by **cytosolic glycerol-3-phosphate dehydrogenase** in order to convert the **DHAP** to **glycerol-3-phosphate**
- The electrons are then transferred to **FAD** by **mitochondrial glycerol-3-phosphate dehydrogenase** when glycerol-3-phosphate is converted back to DHAP
- The **FADH₂** can then transfer these electrons to **coenzyme Q** (**ubiquinone**) via complex II in the electron transport chain.

▼ What is the role of the malate-aspartate shuttle?

- Its role is to **transfer NADH (electrons)** from the **cytosol into the mitochondrial matrix** so that it can be **reoxidised in oxidative phosphorylation**
- It does this for the **liver, kidney and heart**

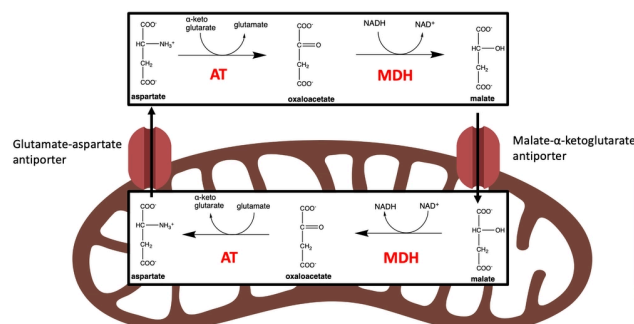
▼ How can oxaloacetate be converted to malate for the malate-aspartate shuttle?

- **Reduction** via the **addition of two hydrogen atoms (H₂)**

▼ Describe the function of the malate-aspartate shuttle.

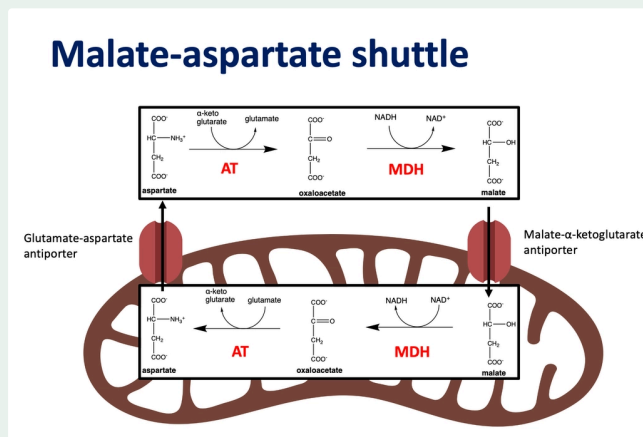
- Aspartate reacts with alpha-ketoglutarate in a transamination reaction to form **oxaloacetate** and **glutamate** via aspartate transaminase
- **Oxaloacetate** is reduced to **malate** via **malate dehydrogenase**
 - During this reaction, the NADH is oxidised to form NAD⁺ and the **electrons are transferred to the malate**
- **Malate** takes the electrons through the **malate-alpha-ketoglutarate antiporter** whilst **alpha-ketoglutarate** moves in the opposite direction
- **Malate** is then oxidised back to **oxaloacetate** via **malate dehydrogenase** and **NAD⁺** is reduced again to form **NADH**
 - **Oxaloacetate** is then reacted with **glutamate** to form **aspartate** and **alpha-ketoglutarate** via **aspartate transferase**
 - The **aspartate** travels through the **aspartate-glutamate antiporter** (while glutamate goes the other way) and another cycle can begin.

Malate-aspartate shuttle



- ▼ What are antiporters and what are the names of the two present in the malate-aspartate shuttle?
- They allow two molecules to travel in opposite directions through them
 - Glutamate-aspartate
 - Malate- α -ketoglutarate
 - This is necessary because the movement of these molecules are effectively what facilitates the transfer of the NADH electrons from the cytosol to the mitochondrial matrix

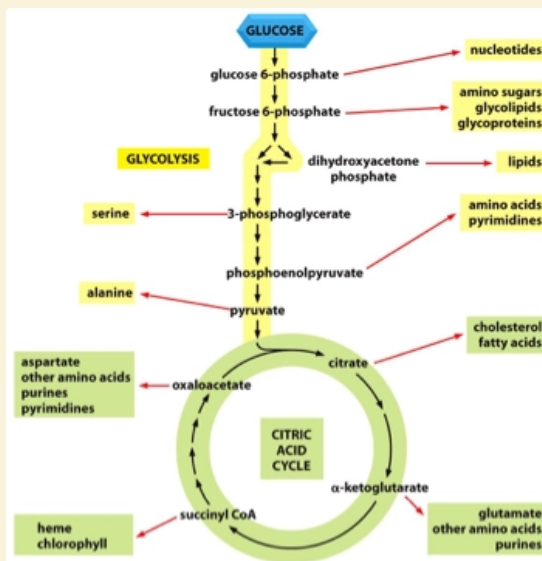
- ▼ Which enzyme in the malate-aspartate shuttle is responsible for the oxidation of NADH in the cytosol and its subsequent reduction in the mitochondria?
- Malate dehydrogenase (MDH)



- ▼ What is reoxidation of cofactors (what happens)?
- This is when cofactors lose the electrons they have gained previously (from glycolysis/Krebs Cycle)
- ▼ How many ATP molecules are formed by the reoxidation of NADH?
- 3
- ▼ How many ATP molecules are formed by the reoxidation of FADH_2 ?
- 2

▼ How else can the intermediates of glycolysis and the TCA Cycle help individuals?

- They can be used to provide building blocks for biosynthetic reactions.
- Formation of **cholesterol from acetyl CoA**
- Formation of **nucleotides from glucose-6-phosphate** via the **pentose phosphate pathway**
- Formation of **amino acids from ketoacids (reverse of original transamination)**



▼ Describe the effects that TCA Cycle enzyme defects can have on individuals.

- Malfunction of enzymes involved in the TCA Cycle can cause certain intermediates to not be produced, and hence the cycle won't be run through correctly
- The cell will hence choose to switch to glycolysis as the main form of ATP production

▼ What is the name of the shuttle used to transport fatty acids from the cytoplasm to the mitochondria of cells for metabolism?

- **Carnitine shuttle**

▼ Describe the structure of a fatty acid.

- It is a straight-chain hydrocarbon which has a -COOH group on the end of the molecule.

- ▼ Diagram showing the beta-oxidation cycle.

The β -oxidation cycle

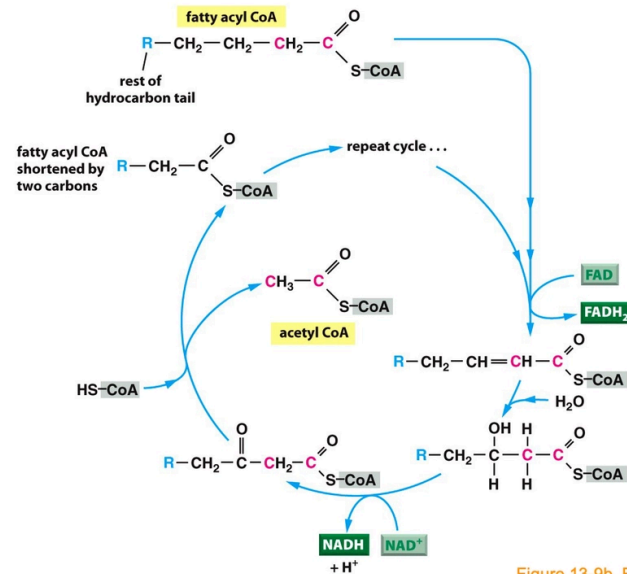


Figure 13-9b Essential Cell

- ▼ Describe how fatty acids are prepared for the beta-oxidation cycle.
- Initially, **fatty acids** are converted to **acyl CoA** species by **acyl CoA synthetase** enzymes
 - NOTE: Acyl CoA species (i.e acetyl CoA) refer to these acyl groups on fatty acids attached to Coenzyme A
 - They do this through the action of **acyl CoA synthetase** and use energy from **2 high-energy ATP bonds**
 - This can come from either the breakdown of ATP to either AMP in one ATP molecule, or ATP to ADP in two molecules
- ▼ Which type of molecule enters the beta-oxidation cycle?
- Acyl CoA species**
 - This is **after they have been converted from their acyl (fatty acid) forms by acyl CoA synthetase**

- ▼ Describe what occurs in the beta-oxidation cycle for fatty acids.
- During the beta-oxidation cycle, a series of oxidation + hydrolysis, oxidation and thiolysis (reaction with HS-CoA to shorten the chain) reactions occur to the original acyl CoA molecule
 - During this process **1x FADH₂** and **1x NADH** are produced
 - The **acetyl CoA** species can also be used to **produce more ATP** through the Krebs Cycle
 - Overall, acyl CoA's (derived from fatty acids) are broken down to acetyl CoA (2Cs) and a shortened acyl CoA (by 2Cs as it has lost acetyl CoA) which can be used in another beta oxidation cycle, whilst producing 1x NADH and 1x FADH₂ in the process

The β-oxidation cycle

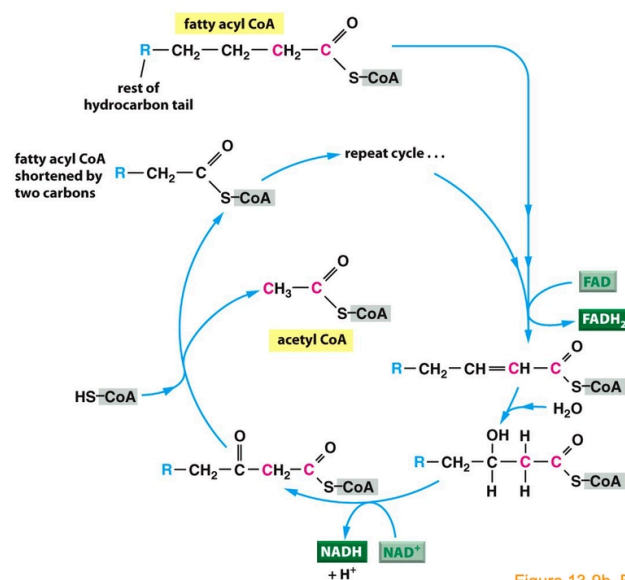
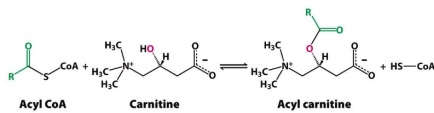


Figure 13-9b Essential Cell

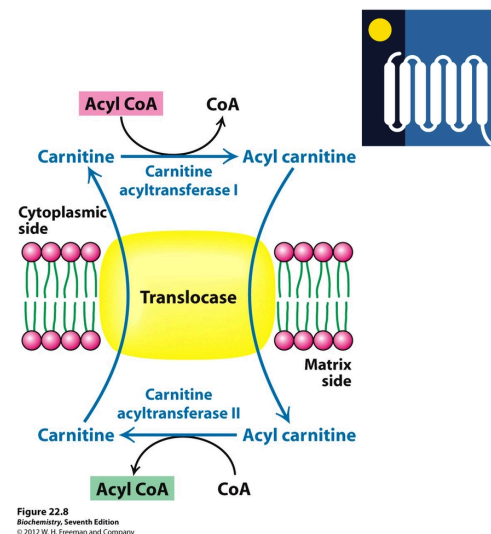
▼ Describe the role and function of the carnitine shuttle.

- **Acyl CoA** is generated in the first step of fatty acid **beta-oxidation**
- Its role is to **transport acyl CoA** from the cytosol to the mitochondrial matrix, where it can be metabolised
- It uses enzymes called **carnitine acyltransferases** to convert **carnitine to acyl carnitine outside** and **acyl carnitine to carnitine inside** so that the acyl group can be transported into the matrix by a **translocase**
- The **acyl group carried by the acyl carnitine** is then **added to CoA** in the matrix to reform the **acyl CoA**

The carnitine shuttle



- Generation of the Acyl CoA species occurs on the outer mitochondrial membrane.
- To transport the species into the matrix it is coupled to the molecule carnitine to form acyl carnitine.
- Carnitine and Acyl carnitine are moved to and from the matrix by a translocase.



▼ What type of genetic disorder is primary carnitine deficiency?

- **Autosomal recessive**
- **Mutations in a gene** known as SLC22A5 which **encodes a carnitine transporter** result in reduced ability of cells to take up carnitine, needed for the β -oxidation of fatty acids.

▼ How does primary carnitine deficiency affect individuals?

- The transporter deficiency means that less acyl CoA is transported into the mitochondrial matrix as less acyl carnitine goes through the carnitine shuttle
- This ultimately means less fatty acids are metabolised as they cannot be metabolised in the cytosol.

▼ How can a primary carnitine deficiency be treated?

- It can be **remedied using carnitine supplements**.

▼ What type of molecule enters the beta-oxidation cycle?

- **Acyl CoA**

▼ What are the products of one beta oxidation cycle?

- 1x acetyl CoA (2C)
- 1x shortened acyl CoA (minus 2C)
- 1x NADH
- 1x FADH₂

▼ Describe what occurs when palmitic acid (16C) is beta-oxidised.

- It is converted to palmitoyl CoA (16C)
- It goes through the beta-oxidation cycle 7 times to produce **8 acetyl-CoA molecules (2C), 7 NADH and 7 FADH₂**
- It produces 8 acetyl-CoA molecules because the last cycle produces 2x acetyl CoA molecules instead of 1x acetyl CoA and 1x shortened acyl CoA
- These can be used to produce ATP, and A LOT is produced.

▼ When and why are ketone bodies formed from acetyl CoA?

- Ketone body formation is an **alternative fate for acetyl CoA** that usually occurs during the **fasting state**
- This is because the **TCA Cycle intermediates (such as oxaloacetate)** are usually **committed to gluconeogenesis (production of energy when fasting)** and so **acetyl CoA won't be able to enter the TCA Cycle** under these conditions.
 - Acetyl CoA needs to combine with oxaloacetate when it enters the TCA Cycle

▼ Describe briefly what lipogenesis is.

- It is the formation of a fatty acid.
- Effectively the reversal of beta oxidation (fatty acid metabolism/breakdown)

▼ What are the two molecules involved in lipogenesis?

- **Malonyl CoA (3C) and acetyl CoA (2C)**
- These molecules are used as the **building blocks of lipogenesis**
- **Acyl carrier proteins** are used to enable the continuation of lipogenesis - **they act as a base upon which more fatty acids can be added**

▼ How many enzymes are involved in lipogenesis and what are their names?

- 2.
- **Acetyl CoA Carboxylase**
- **Fatty acid synthase**

▼ Describe three differences between the synthesis and degradation of lipids.
IP